Excel Functions for Data Exploration

Practical, visual guide using the Supermarket dataset

COUNT

SUM

COUNTIF

IF

RANK

CORREL

Learning Objectives

Choose functions based on the question you want to answer.

- Count records, non-empty cells and unique values
- Calculate totals, averages, medians, minimums and maximums
- Use COUNTIF / COUNTIFS / SUMIF / SUMIFS for segmented analysis
- Use IF, AND and OR to classify transactions
- Find top/bottom values with RANK, LARGE and SMALL
- Validate data with ISBLANK, ISNUMBER, ISTEXT and IFERROR
- Work with dates and calculate Shipping Days
- Measure variation and relationships with STDEV.S, VAR.S and CORREL

Data Exploration Function Map

A progression from basic structure to deeper analysis.

1. Structure

COUNT

COUNTA

COUNTBLANK

UNIQUE

2. Summary

SUM

AVERAGE

MEDIAN

MIN / MAX

MODE.SNGL

3. Conditions

COUNTIF / COUNTIFS

SUMIF / SUMIFS

AVERAGEIF / AVERAGEIFS

4. Logic & Quality

IF

AND / OR

ISBLANK / ISNUMBER

IFERROR

5. Advanced

RANK.EQ

LARGE / SMALL

PERCENTILE.INC

STDEV.S / VAR.S

CORREL

1. COUNT Family

Understand dataset size, completeness and distinct values.

COUNT

=COUNT(R2:R9801)

Counts numeric Sales values.

COUNTA

=COUNTA(B2:B9801)

Counts non-empty Order IDs.

COUNTBLANK

=COUNTBLANK(L2:L9801)

Counts missing Postal Codes.

UNIQUE

=UNIQUE(M2:M9801)

Returns unique Region values.

=COUNTA(UNIQUE(B2:B9801))

Count distinct Order IDs in modern Excel.

2. Descriptive Statistics

Turn Sales into a numerical profile.

SUM

=SUM(R2:R9801)

Total Sales

AVERAGE

=AVERAGE(R2:R9801)

Mean Sales

MEDIAN

=MEDIAN(R2:R9801)

Middle value

MIN / MAX

=MIN(R2:R9801) / =MAX(R2:R9801)

Range endpoints

MODE.SNGL

=MODE.SNGL(R2:R9801)

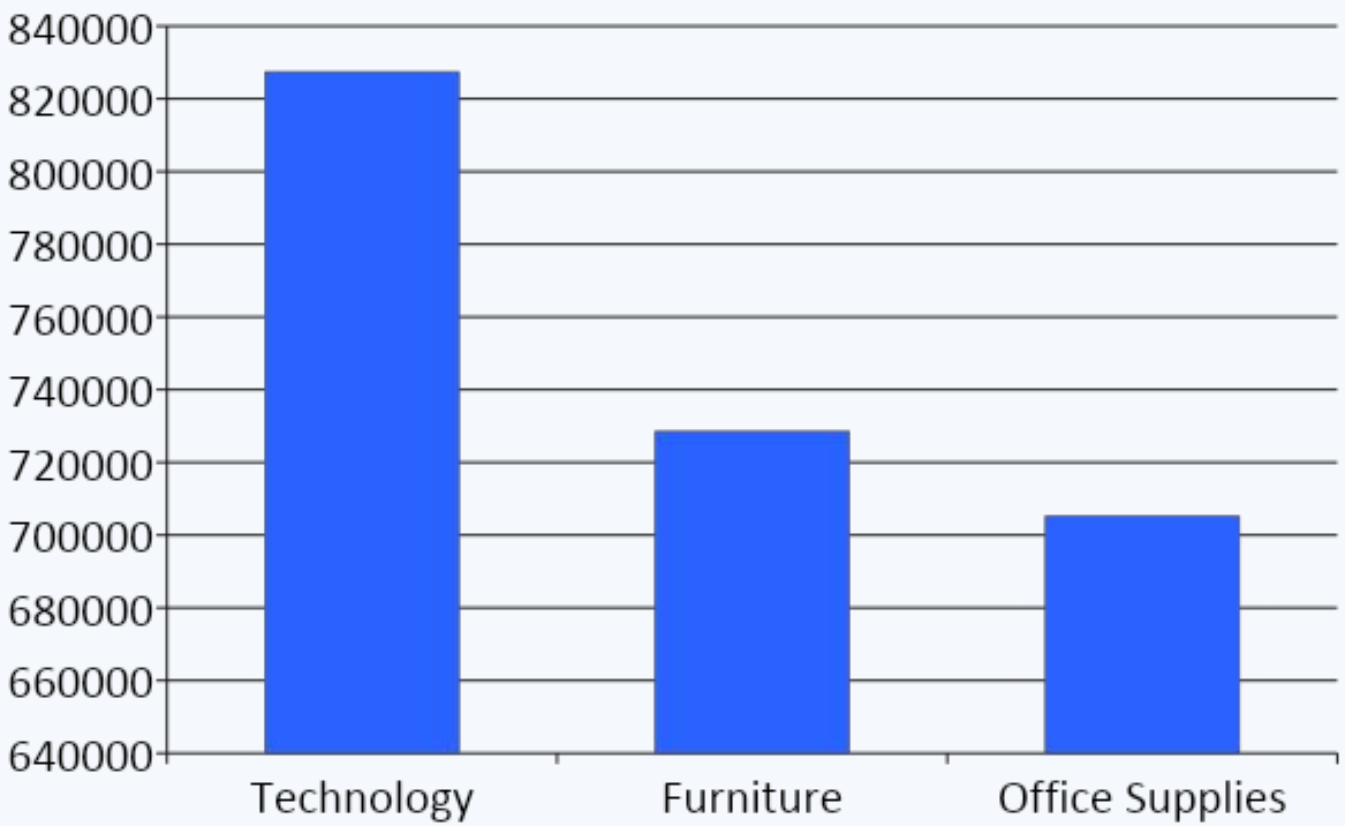
Most frequent value

```
=MAX(R2:R9801)-MIN(R2:R9801)
```

Range = maximum − minimum.

Figure: Sales by Category

Build summaries with SUMIF, then visualize them.



SUMIF examples

Technology:

=SUMIF(O:O,"Technology",R:R)

Furniture:

=SUMIF(O:O,"Furniture",R:R)

Office Supplies:

=SUMIF(O:O,"Office Supplies",R:R)

3. Conditional Aggregation

Answer “how many?”, “how much?” and “what is the average?” under conditions.

COUNTIF

=COUNTIF(O:O,"Technology")

Rows in Technology

COUNTIFS

=COUNTIFS(M:M,"West",O:O,"Technology")

West + Technology

SUMIF

=SUMIF(O:O,"Technology",R:R)

Technology Sales

SUMIFS

=SUMIFS(R:R,M:M,"West",O:O,"Technology")

West Technology Sales

AVERAGEIF

=AVERAGEIF(O:O,"Technology",R:R)

Average Technology Sales

AVERAGEIFS

=AVERAGEIFS(R:R,M:M,"West",O:O,"Technology")

Average West Technology Sales

4. IF — Turn Numbers into Business Labels

IF converts a condition into a category or decision flag.

```
=IF(R2>=500,"High Sales","Regular Sales")
```

Syntax: =IF(logical_test, value_if_true, value_if_false)

High / Low Sales

=IF(R2>=500,"High","Low")

Use for transaction segmentation.

Shipping Class

=IF(S2<=2,"Fast","Standard")

Use after calculating Shipping Days.

Missing Data Flag

=IF(L2="","Missing","Available")

Useful for data-quality checks.

5. AND, OR and Nested IF

Combine multiple conditions.

```
=IF(AND(R2>=500,S2<=3),"Priority Order","Normal Order")
```

AND: every condition must be TRUE.

```
=IF(OR(O2="Technology",O2="Furniture"),"Core Category","Other")
```

OR: at least one condition must be TRUE.

```
=IF(R2>=1000,"Very High",IF(R2>=500,"High",IF(R2>=100,"Medium","Low")))
```

Nested IF: multiple categories.

≥1000 → Very High

500–999 → High

100–499 → Medium

<100 → Low

6. Ranking, Top and Bottom Values

Identify high-value transactions and values for investigation.

RANK.EQ

=RANK.EQ(R2,$R$2:$R$9801,0)

1 = highest

LARGE

=LARGE(R2:R9801,1)

Largest value

SMALL

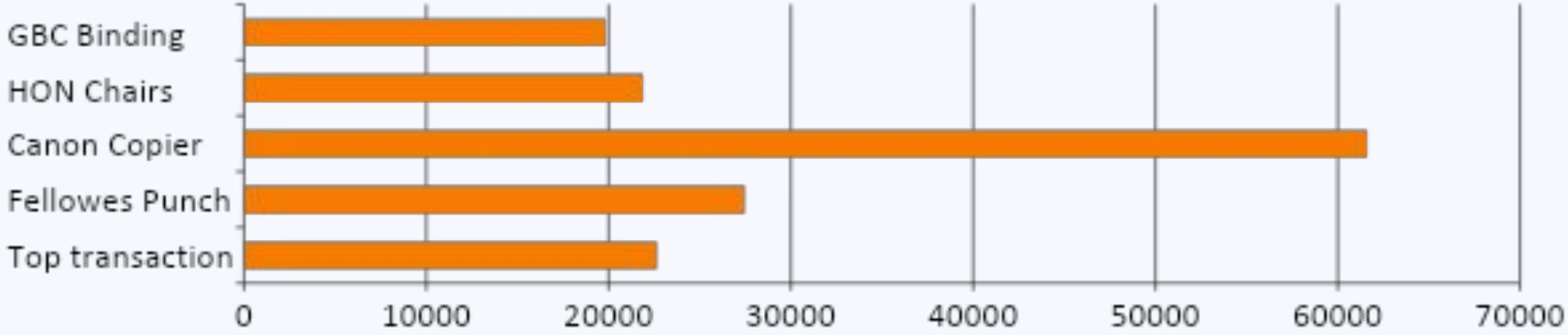
=SMALL(R2:R9801,1)

Smallest value

PERCENTILE

=PERCENTILE.INC(R2:R9801,0.95)

95th percentile



7. Data Quality Functions

Check blanks, data types and formula errors before analysis.

ISBLANK

=ISBLANK(L2)

Is Postal Code missing?

ISNUMBER

=ISNUMBER(R2)

Is Sales numeric?

ISTEXT

=ISTEXT(B2)

Is Order ID text?

IFERROR

=IFERROR(formula,"Check")

Handle errors cleanly.

```
=IF(ISNUMBER(R2),"Valid Sales","Check Sales")
```

Combine validation functions with IF.

8. Date Functions and Shipping Days

Dates support time trends and delivery analysis.

YEAR

=YEAR(P2)

Extract year

MONTH

=MONTH(P2)

Extract month

DAY

=DAY(P2)

Extract day

TEXT

=TEXT(P2,"mmm")

Month label

=Q2-P2

Shipping Days = Ship Date − Order Date.

=IF(S2<=2,"Fast",IF(S2<=4,"Standard","Slow"))

Classify delivery speed after Shipping Days is calculated.

9. Variation and Relationship

Go beyond averages: understand spread and linear association.

STDEV.S

=STDEV.S(R2:R9801)

Measures sample spread.

VAR.S

=VAR.S(R2:R9801)

Variance is squared standard deviation.

CORREL

=CORREL(range1,range2)

Linear association: −1 to +1.

−1
Opposite direction

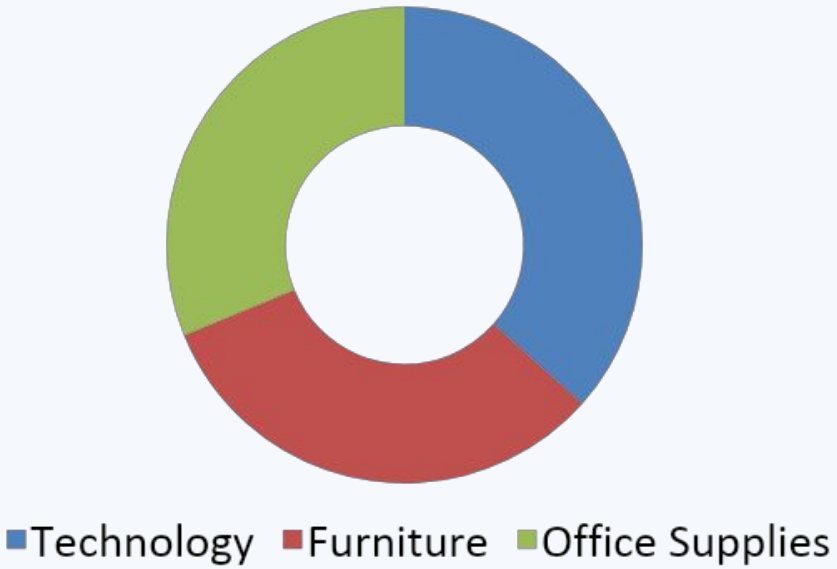
0
No linear association

+1
Same direction

Mini Excel Exploration Dashboard

A compact example of combining formulas and figures.

Total Sales	Rows	Unique Orders	Missing Postal Codes
$2.26M	9,800	4,922	11





Classroom Activity — Explore the Supermarket Data

Complete these tasks in Excel.

1. Count total rows and non-empty Order IDs.
2. Count unique Order IDs and Customer IDs.
3. Find total, average, median, minimum and maximum Sales.
4. Count rows for each Category using COUNTIF.
5. Calculate Sales by Category using SUMIF.
6. Create a High/Low Sales flag using IF.
7. Find the top 10 Sales values using LARGE.
8. Check missing Postal Codes using COUNTBLANK.
9. Calculate Shipping Days and classify Fast / Standard / Slow.
10. Create one chart from your summary table.

From Function → Question → Insight

The purpose is to answer a business question, not simply use a formula.

COUNT / COUNTA	How large is the dataset?	Data volume
COUNTBLANK	Where is data missing?	Data quality
SUM / SUMIF	Where are sales concentrated?	Contribution
AVERAGE / MEDIAN	What is typical?	Central tendency
COUNTIFS / SUMIFS	What happens under multiple conditions?	Segmented analysis
IF / AND / OR	How can records be classified?	Business rules
RANK / LARGE	Which values need attention?	Top-value investigation
STDEV.S / CORREL	How variable / related?	Deeper analysis

Excel Data Exploration Cheat Sheet

Quick reference for the lab.

Counting

COUNT • COUNTA • COUNTBLANK • UNIQUE

Summary

SUM • AVERAGE • MEDIAN • MIN • MAX • MODE.SNGL

Conditional

COUNTIF • COUNTIFS • SUMIF • SUMIFS • AVERAGEIF • AVERAGEIFS

Logic

IF • AND • OR

Ranking

RANK.EQ • LARGE • SMALL • PERCENTILE.INC

Quality

ISBLANK • ISNUMBER • ISTEXT • IFERROR

Dates

YEAR • MONTH • DAY • TEXT • date subtraction

Statistics

STDEV.S • VAR.S • CORREL

Key Takeaway

Explore → Summarize → Compare → Validate → Classify → Interpret

Choose the function that matches the business question.